

Gruhesh Sri Sai Karthik Kurra

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huggingface.co/karthik-2905 · kaggle.com/gruheshkurra · orcid.org/0009-0002-0558-2882

Incoming MSc Computing (Artificial Intelligence and Machine Learning) student at Imperial College London; first author of an accepted IEEE ICCMMLA 2025 paper on morphology-aware embeddings and of a 3.9M-parameter language model written in pure NumPy with its own autograd. Seeking a research engineering internship in model training, evaluation, or efficient inference.

EDUCATION

Imperial College London, United Kingdom

Sep 2026 – Sep 2027 (incoming)

MSc Computing (Artificial Intelligence and Machine Learning), Department of Computing

KL University, Hyderabad, India

Aug 2021 – May 2025

B.Tech Computer Science and Engineering · CGPA 9.72 / 10

- Implemented transformer, diffusion, GAN, and graph neural network models from their equations in NumPy and PyTorch; public write-ups at blogs.gruheshkurra.com
- IELTS Academic overall 7.5 (CEFR C1)

RESEARCH EXPERIENCE

Research Intern

Dec 2024 – Jun 2025

IIIT Hyderabad · Hyderabad, India

- Designed a layout-preserving document translation system: Vision Transformer layout detection (92% accuracy), character-metric font detection, and K-means colour-palette extraction (88% accuracy)
- Built the FastAPI + Tesseract optical character recognition (OCR) pipeline for English, Hindi, Telugu, and German that keeps source fonts, colours, and page structure in the translated output
- Presented a live demo at the IIIT Hyderabad research expo (May 2025) to faculty and industry visitors; demo video: youtu.be/rcvSuBcBjyg

AI & ML Research Trainee

May 2024 – Oct 2024

iHub-Data, IIIT Hyderabad · Hyderabad, India

- Completed a six-month faculty-mentored research programme covering architecture design, LLM fundamentals, prompting, fine-tuning, quantisation, and deployment
- Applied OCR and document-vision methods that were later reused in the layout-translation project

PUBLICATIONS

Accepted

Morphology-Aware Nested Character Embeddings for Word Representation (MANCE)

Gruhesh Sri Sai Karthik Kurra, Chandanasree Moparathi, Prathipati Manish Chowdary, Pavan Kumar Pagadala

Accepted, IEEE ICCMMLA 2025. [IEEE Xplore 11580466](https://arxiv.org/abs/2504.11580) · github.com/GruheshKurra/MANCE-NLP

- Hierarchical Transformer, CNN, and BiLSTM encoders over nested character sequences; morphological-variant sentiment 91.67% vs 33.33% for Word2Vec (+58.3 points); DBpedia 99.0%, AG News 93.5%, GermEval 2014 NER 81.3% F1
- Morphology breakdown: 87.3% plurals, 85.9% past tense, 81.5% superlatives, against 75.6% for FastText; character-level vocabulary of fewer than 100 symbols; paper reports 2.2× faster training than the word-level baseline

Under review

Dual-Stream Artifact Detection with Iterative Evidence Refinement for Frame-Level Deepfake Recognition

Gruhesh Sri Sai Karthik Kurra, Pavan Kumar Pagadala, Malathy Batumalay, Veda Boddapati

Under review, IEEE NEPCON 2026 (submitted 2026). github.com/GruheshKurra/radar_deepfake · Kaggle frame corpus

- Boundary (Sobel-edge) and frequency (multi-band FFT) streams trained under an orthogonality penalty and fused by an iterative cross-attention Evidence Refinement Module; trained on 227,504 frames from FaceForensics++, Celeb-DF v2, and WildDeepfake
- Ran a video-disjoint leakage audit of WildDeepfake and measured 96.3% AUC overall on the corrected split (87.1% on Celeb-DF); released the code, frame corpus, audit script, and trained checkpoint

Prism Tuning: Repulsion-Trained Seed Embeddings in a Frozen Transformer for Non-Redundant Generation

Gruhesh Sri Sai Karthik Kurra, Pavan Kumar Pagadala, Malathy Batumalay, Sushma Reddy Koduru

Under review, IEEE NEPCON 2026 (submitted 2026)

- K learnable seed vectors trained by output-level repulsion inside one frozen transformer, giving K non-redundant generation streams from a single prompt; diversity is paid once at training with zero inference overhead, unlike generate-then-deduplicate
- Uses a generated-position hidden-state surrogate for non-differentiable decoded text, corrects representation anisotropy in the repulsion loss, and splits each seed into shared-style and repelled-content parts; method paper, no empirical results claimed
- Grew out of the Zynthetix parent/child synthetic-data architecture, where the same non-redundancy idea was first designed

Preprints (open research with DOI, not peer-reviewed)

BIE: Bit-Index Encoding for Efficient Neural Network Weight Compression

Gruhesh Sri Sai Karthik Kurra

Zenodo, 28 Sep 2025, DOI 10.5281/zenodo.17217218. zenodo.org/records/17217218 · code (MIT)

- Weight compression of up to $40\times$ at reconstruction MSE below 10^{-6} ; Numba JIT sparse matmul kernel, strongest against baselines at sparsity above 70%

Hybrid RAG-Enhanced Deepfake Detection: Combining Retrieval-Augmented Generation with Visual Inconsistency Analysis

Gruhesh Sri Sai Karthik Kurra

Zenodo, Aug 2024. zenodo.org/records/16732053

- Privacy-first detector: retrieval-augmented generation (RAG) over CLIP embeddings with FAISS, visual inconsistency analysis, and an uncertainty-aware classifier, processed locally rather than cloud-only; 94.8% accuracy, 94.6% F1

RADAR: Reasoning-Augmented Deepfake Artifact Recognition via Multi-Branch Evidence Aggregation

Gruhesh Sri Sai Karthik Kurra

Preprint, 2026

- Deepfake detection as structured evidence aggregation: three specialised branches (skin texture, boundary aliasing, AI-generation fingerprints) feed an iterative cross-attention reasoning module (R-Former) that outputs per-artifact confidence scores and saliency maps; trained from random initialisation to avoid ImageNet bias
- Architecture and evaluation protocol only; empirical results are reported in the Dual-Stream companion paper

Journal articles (IJNRD)

- “Voice-Activated AI for Seamless Computer Interaction.” IJNRD, Jan 2025. Voice command system with intent classification and system control
- “Global Remote RAM Sharing: A Novel Framework for Distributed Computational Systems.” IJNRD, Dec 2024
- “Dynamic Auto-Finetuning of Language Models Based on Confidence-Driven Knowledge Integration.” IJNRD, Nov 2024. Confidence scores trigger selective fine-tuning without a full retrain

MACHINE LEARNING PROJECTS

From-Scratch NumPy Language Model (AL-1 A) github.com/GruheshKurra/core-language-model · HF [model-a-scratch](#)

- Wrote a **3,869,184-parameter** decoder-only language model in pure NumPy with a custom autograd engine: 4 layers, $d_{\text{model}} 256$, 8 query heads with 2 key-value heads (grouped-query attention, GQA), SwiGLU, rotary position embeddings (RoPE), QK-Norm, pre-LN RMSNorm, tied embeddings, 256-token context, and a KV-cache for generation
- Built a byte-level BPE tokeniser (4,096 vocabulary, 3,835 merges, chat-role specials); pretrained on DailyDialog (1.62M training tokens, 2,000 steps at batch 32×256 , AdamW) then supervised fine-tuned on 19,375 EmpatheticDialogues conversations with assistant-only loss for 800 steps
- 78 unit tests across 12 files cover the autograd, model, tokeniser, and training code

Qwen3-0.6B Tool-Calling Fine-Tune (AL-1 B) github.com/GruheshKurra/AL1-model-B · HF [AL1-model-B](#)

- Fine-tuned Qwen3-0.6B with LoRA supervised fine-tuning (rank 16, $\alpha=32$, dropout 0.05, all attention and MLP projections) using TRL SFTTrainer with assistant-only loss on 1,423 training examples (945 tool, 478 chat) for 3 epochs on a RunPod RTX A6000
- Taught five file-system and shell tools in the Hermes tool-call format; on a 12-case greedy eval, tool-name accuracy rose $0.50 \rightarrow 1.00$ and full-call exact match $0.50 \rightarrow 0.917$ (11/12) against the base model

DeepGuard: On-Device Deepfake Detection github.com/GruheshKurra/Deepguard · demo

- Trained an EfficientNet-B1 classifier on 25k images to **98.62%** accuracy in 55 minutes on an Apple M4; converted it to a **26 MB** Core ML model for iOS that runs in under 500 ms per image on device

GPT-2 From Scratch

github.com/GruheshKurra/FirstGPTFromScratch

- Implemented the 124M-parameter GPT-2 architecture in PyTorch with a byte-pair-encoding tokeniser and a full training loop, trained locally; full walkthrough published as post 13 of the AI from Scratch series

Transformers From Scratch

github.com/GruheshKurra/TransformersFromScratch

- Replicated the encoder-decoder Transformer of *Attention Is All You Need* (multi-head attention, positional encodings, masking) in PyTorch with BLEU benchmarks

NL2SQL Pretrained Model

github.com/GruheshKurra/nl2sql-pretrained

- Fine-tuned an attention-based language model to translate plain-English questions into SQL, reaching perplexity 1.42

Clinical Trial Similarity Analysis

github.com/GruheshKurra/Clinical-Trial-Similarity-Analysis

- Built a retrieval system over clinical trial descriptions with PubMedBERT embeddings and a FAISS index; 99.5% similarity accuracy (self-reported)

LLaMA-Style Training Recipe

github.com/GruheshKurra/LLamaModel

- Implemented a LLaMA-style decoder (RMSNorm, RoPE, SwiGLU) with a training recipe in PyTorch

ADDITIONAL PROJECTS

- **AI Voice Assistant for Windows:** voice-activated computer control with intent classification. [AI-Voice-Assistant-for-Windows](#)
- **WSO2 on Kubernetes:** deployment support for WSO2 Micro Integrator. [WSO2-Kubernetes-Support](#)
- **Domain applications:** FarmCare (agriculture), healthcareAI (public); SeismoCare (private)
- **NLP learning roadmap:** curated study path for NLP and LLMs. [awesome-ai-roadmaps](#)
- **Early demos:** hand-gesture mouse control, StoryMaker, FastAPI chat

INDUSTRY EXPERIENCE

AI Integration Engineer

Oct 2025 – Nov 2025

InfoAjax Consulting Pvt Ltd · Remote, Hyderabad

- Built an integration layer between enterprise applications and cybersecurity services over standardised APIs, using Azure OpenAI agents for schema mapping and connection setup learned from prior runs
- Connected Android and Apple enterprise applications to Azure services, and delivered several proofs of concept linking an Azure-hosted enterprise application to Salesforce
- Implemented a React 18 + FastAPI + WebSocket console for live agent monitoring; wired Azure AD through Microsoft Graph for application discovery and metadata
- Prototyped a screen-driven automation agent that captured screenshots in a loop and issued click and type actions from them; proof of concept only, and not reliable enough in real time to connect to live systems

WSO2 API Developer

Oct 2024 – Jun 2025

InfoAjax Consulting Pvt Ltd · Remote, Hyderabad

- Shipped RESTful ticketing APIs for PLDT (Philippines telecom) in WSO2 Integration Studio (Eclipse-based), working with an international client under SLAs
- Managed production secrets through GitLab vault and deployed integrations on WSO2 Choreo; debugged production issues against sparse vendor documentation
- Related public work: [WSO2-Kubernetes-Support](#)

Founder & CEO

Mar 2024 – Jan 2025

Zynthetix · Hyderabad, India

- Designed the architecture for a privacy-preserving synthetic-data system: a meta-controller in which a parent model coordinates hundreds of specialised child models to generate tabular, image, and text data without duplication
- The same non-redundancy principle became the basis of the Prism Tuning paper (repulsion-trained seed embeddings)

LEADERSHIP

Technical Lead

2023 – 2025

Student Activity Council, KL University · SAC Momentum Award

- Mentored 20+ juniors in web development and ML
- Ran technical workshops for 100+ students

- Coordinated 15+ technical events across the council calendar

Hackathon Mentor

TBC year

Anurag University 24-hour hackathon · Hyderabad, India

- Conducted the 24-hour hackathon for 45 teams of three, answered technical questions throughout, and helped teams carry their projects to completion

AWARDS

- **1st Place, Global Challenge Lab 2026**, Imperial College London (Jul 2026): Imperial Enterprise Lab and Futurize global innovation sprint, 1,000+ students, 14 days, four tracks, one Demo Day; Team Corio (five people across AI/ML, aerospace, physics and mechanical, finance) won with OrbitOps, a vendor-neutral reliability layer for satellite onboard AI (pre-launch fault injection, in-orbit known-answer probes that catch silent degradation, rollback to a clean copy); led the AI/ML work and technical strategy; £1,500 team prize
- **SAC Momentum Award**, Student Activity Council, KL University (see Leadership)
- **1st Place, University Webathon**, KL University (2022): diet-management platform with personalised meal planning, calorie tracking, and nutritional recommendations, built in a 4-hour hackathon against 50+ teams
- **2nd Place, Design Expo**, KL University (2022–2023): Arduino smart switchboard for IoT home automation with real-time energy monitoring and mobile-app control

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Swift, TypeScript, JavaScript, SQL, Shell (Bash), R, PHP, HTML, CSS

Deep learning: PyTorch, TensorFlow, Keras, NumPy (own autograd engine), Hugging Face Transformers, TRL, PEFT / LoRA, PyTorch Geometric, Gradio, Streamlit

Classical ML and data science: scikit-learn, pandas, NumPy, Matplotlib, UMAP, statistical modelling, exploratory data analysis, feature engineering, synthetic data

LLMs and generative models: Transformer architectures (attention, GQA, RoPE, SwiGLU, RMSNorm, KV-cache), byte-level BPE tokenisation, supervised and small-language-model fine-tuning, tool-calling SFT, prompt engineering, RAG, LangChain, LlamaIndex, Azure OpenAI agents, GANs, VAEs, diffusion models (DDPM), GNNs

Computer vision and document AI: EfficientNet, Vision Transformers, CLIP, OpenCV, MediaPipe, deepfake and artifact detection (Sobel / FFT streams), Tesseract OCR, document analysis, layout detection, font and colour-palette extraction

Model efficiency: LoRA, quantisation, pruning, distillation, weight compression (Bit-Index Encoding), Numba JIT kernels, on-device inference (Core ML, Apple Silicon / Metal MPS)

iOS and mobile: Swift, SwiftUI, Core ML, Vision framework, AVFoundation, Core Data, React Native

Web and backend: FastAPI, REST, WebSocket, React 18, Next.js, Node.js, Express.js, TailwindCSS, Vite, Firebase, Supabase

Integration and enterprise: WSO2 Integration Studio, WSO2 Micro Integrator, WSO2 Choreo, Azure AD, Microsoft Graph, Salesforce integration, Azure services for Android and iOS enterprise apps, Automation Anywhere

Cloud and DevOps: AWS (EC2, S3, Lambda), Azure, Google Cloud, Oracle Cloud Infrastructure, RunPod, Vercel, serverless, Docker, Kubernetes, CI/CD, GitHub Actions, GitLab CI, Git

Data stores: PostgreSQL, MySQL, MongoDB, Redis, FAISS

Hardware and embedded: Arduino, IoT home automation

Research practice: experiment design, leakage audits and video-disjoint splits, ablations, IEEE manuscript preparation, Zenodo / DOI releases, Kaggle dataset publishing, technical writing (15-post AI from Scratch series)

CERTIFICATIONS

- Certificate of Achievement, Global Challenge Lab 2026 (1st Place), Imperial College London (2026)
- Become a Linear Algebra Master, Udemy (2026)
- A Deep Understanding of Deep Learning (with Python), Udemy (2026)
- WSO2 Certified Micro Integrator Developer – V4, WSO2 (2025)
- WSO2 Certified Micro Integrator Practitioner – V4, WSO2 (2025)
- Google Developer Groups certificate, GDG (2025)
- TensorFlow Developer Certificate, Google (2024)
- Oracle Cloud Infrastructure Generative AI Professional, Oracle (2024)
- Red Hat Certified Enterprise Application Developer, Red Hat (2024)
- Certified Advanced Automation Professional, Automation Anywhere (2024)
- Build Large Language Models From Scratch, Visuara (2024)
- Hugging Face MCP Course, Unit 1, Hugging Face (2024)
- Hugging Face MCP Course, Unit 3, Hugging Face (2024)
- Six-Month AI/ML Research Training, iHub-Data, IIIT Hyderabad (2024)

- AWS Certified Solutions Architect – Associate, Amazon Web Services (2023)
- AWS Certified Cloud Practitioner, Amazon Web Services (2023)
- Oracle Cloud Infrastructure Architect Associate, Oracle (2023)
- Oracle Database Certificate, Oracle (2023)
- Certificate of Service, Volunteer, Street Cause NGO (2023–24)
- Certificate of Appreciation, Student Activity Center, KL University (2022–23)

WRITING

- **AI from Scratch**, a 15-post series at blogs.gruheshkurra.com/series/ai: linear algebra through a working GPT-2, in order, each post with the derivation, a worked numeric example from the real config, and runnable code
- Covers reverse-mode autograd and gradient checking, byte-pair encoding, embeddings and positional encoding, cross-entropy and AdamW, attention and a full Transformer forward and backward pass by hand, GPT-2 (124M) in PyTorch, DeepSeek V4 long-context engineering, and the 3.87M-parameter NumPy chat model above

INTERESTS

ML-from-scratch writing at blogs.gruheshkurra.com; mentoring student developers through the Student Activity Council and hackathons. Open to collaboration on transformer research, LLM development, on-device AI, vision-language models, neural compression, and agentic systems